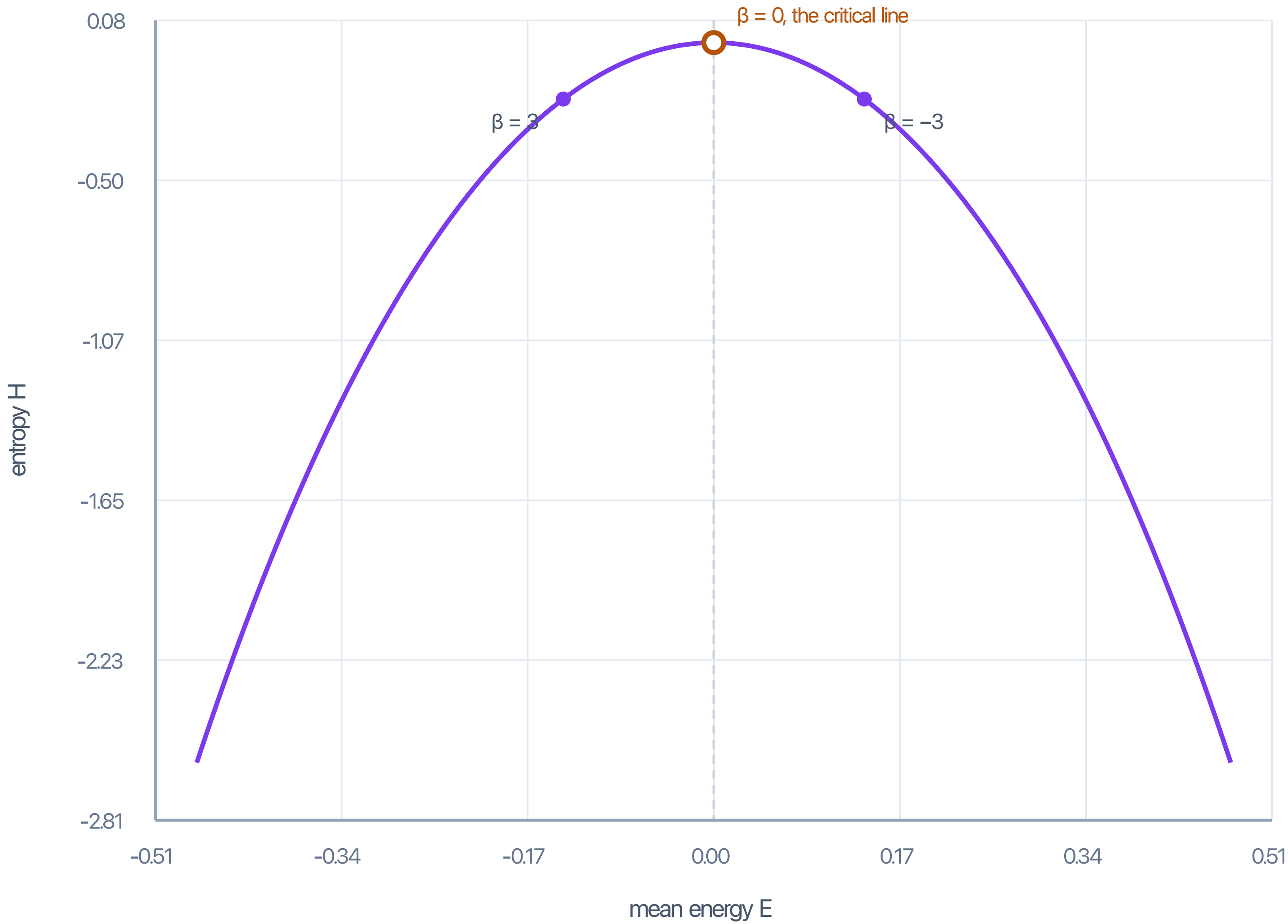
The energy–entropy curve of the completed zeta function

ζ has none — its region is unbounded at the pole. ξ does, for every real β.



Z(β) = ξ(½ + β)

= ∫ Φ(u) e^{βu} du, Φ > 0

Φ even ⟹ Z(β) = Z(−β)

⟹ E odd, H even

the mirror symmetry is

the functional equation

ξ(s) = ξ(1 − s)